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ChE 210 Quiz #3 6 November 2019

6

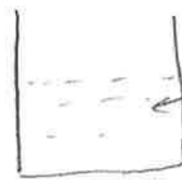
A rigid vessel is initially filled with one mole of a gas mixture of 50 mol% hydrogen (H_2) and 50 mol% acetaldehyde (C_2H_4O) at a total pressure of 760 mmHg absolute and 25˚C. Ethanol (C_2H_6O) forms according to the chemical reaction: $C_2H_4O + H_2 \rightarrow C_2H_6O$

(a) After the system comes to equilibrium at constant temperature of 25˚C, the pressure has dropped to 700 mmHg absolute. Calculate the total moles and the moles of each species present. Assume ideal gas behavior.

(b) If the equilibrium expression is given by $K = \frac{y_{C_2H_4O}\, y_{H_2}}{y_{C_2H_6O}}\left(\frac{P_0}{P(atm)}\right)$ where P_0 = 1 atm and $P(atm)$ is the pressure in atmospheres absolute, what is the value of the equilibrium constant K?

Given,

$C_2H_4O + H_2 \longrightarrow C_2H_6O$



1 mole gas mixture

50 mol% Hydrogen

50 mol% Acetaldehyde

Basis → 1 mole of gas mixture

The volume at 760 mm Hg absolute and 25°C

$PV = nRT$

$V = \frac{nRT}{P} = \frac{(1)\left(\frac{62.361\, mmHg\, L}{gmol\, K}\right)(298\, K)}{760\, mmHg} = 24.45\, L/gmol$

The system comes to equilibrium

$P = 700$ mm Hg

$T = 25°C = 298$ K

Assuming ideal gas behavior,

$PV = nRT$

$n = \frac{PV}{RT} = \frac{(700\, mmHg)(24.45\, L/gmol)}{\left(62.361\, \frac{mmHg\, L}{gmol \cdot K}\right)(298\, K)}$

$n = 0.921$ moles

∴ The total moles = 0.921 moles

The mol % Hydrogen = 50 mol% of 0.921 moles ✗ 0.4605 moles of H_2

The mol % Oxygen = 50 mol % of 0.921 moles ✗ 0.4605 moles of O_2

moles of C_2H_6O = 1 − (0.921 moles)

= 0.079 moles C_2H_6O

where are your mole balance equations?

Don't guess!

b) The equilibrium constant is given by

$$K = \frac{y_{C_2H_4O}\, y_{H_2}}{y_{C_2H_6O}} \left(\frac{P_0}{P\,atm}\right)$$

Here, $P_0 = 1\,atm$

$$P = \frac{750\ mmHg}{760\ mmHg} = 0.987\ atm$$

$$K = \frac{(0.4605)(0.4605)}{(0.079)} \left(\frac{1\,atm}{0.987\,atm}\right)$$

$$\boxed{K = 2.72}$$

close, but method is wrong!